

Epiplexity: information accessible to a bounded learner

Epiplexity with lightweight learning algorithms

Shannon entropy and Kolmogorov complexity do not explicitly restrict the computational resources of the observer. Finzi et al. (Finzi et al., 2026) introduce *epiplexity* as the structural information in a distribution that can be extracted by a computationally bounded observer.

For a T -time probabilistic model P with program length $|P|$, define

$$P^* = \arg \min_{P \in \mathcal{P}_T} \{|P| + \mathbb{E}_X [-\log P(X)]\}.$$

The corresponding decomposition is

$$\underbrace{\text{MDL}_T(X)}_{\text{total time-bounded information}} = \underbrace{|P^*|}_{S_T(X) - \text{epiplexity}} + \underbrace{\mathbb{E}[-\log P^*(X)]}_{H_T(X) - \text{time-bounded entropy}}.$$

Thus, epiplexity is not simply a property of the dataset: it is defined relative to a computational budget and an admissible class of observers. In practice, neural training procedures provide tractable approximations to this program optimization.

Key reference: Finzi et al., “From Entropy to Epiplexity”, 2026. The paper’s definition is compute-bounded; practical estimators approximate the optimal program through training trajectories.

What we want to test: the observer may matter

The original empirical study demonstrates epiplexity using neural observers and computational sweeps over model size and training duration, including synthetic cellular automata and natural-data experiments (Finzi et al., 2026). Our question is narrower and experimentally controllable:

How does the estimated structural information change when the observer is replaced by a lightweight architecture?

We will keep the data-generating process fixed and vary the observer

$$X \xrightarrow{\text{learning algorithm}} P_T \quad \text{with} \quad \text{MLP vs. CNN.}$$

The comparison is motivated by the compute-bounded definition itself: changing the hypothesis class and its inductive bias changes the set of programs that can be practically realized within a fixed budget. CNNs additionally encode locality, whereas MLPs provide a substantially weaker architectural prior.

This lets us separate two questions that are often conflated: *how much structure is present* and *how much structure is accessible to a particular learner*.

Estimators: one learning trajectory, several codes

We will evaluate several measures on the same training trajectories rather than introducing separate experimental setups for each metric.

Prequential coding. The description length is accumulated from predictions made by successive checkpoints before the corresponding training information is available. In the implementation of Finzi et al., the model-code contribution is estimated from the area of the training-loss curve above the final loss (Finzi et al., 2026):

$$K_{\text{preq}} \approx \sum_t \ell_t \Delta t \quad \text{relative to the final-loss baseline.}$$

Requential coding. Qiu et al. (Qiu et al., 2026) replace coding of the exact observed samples by coding synthetic samples on which teacher and student disagree. The central quantity is the cumulative divergence

$$K_{\text{req}} \propto \sum_t \text{KL}(P_t^{\text{teacher}} \parallel P_t^{\text{student}}).$$

This removes much of the cost associated with the entropy of the particular dataset sequence.

EDL. Donoway et al. (Donoway et al., 2026) define Excess Description Length as the gap between sequential training-code length and the residual code length under the final predictor. It is closely related to learning dynamics, but is a measure for a specified learning algorithm rather than the optimization over all admissible programs defining epiplexity.

Experimental design: lightweight observers

We will use a controlled grid: the dataset and compute budget are fixed, while the observer architecture and its scale are varied.

	MLP	CNN
Parameters	10^3 – 10^6	10^3 – 10^6
Capacity	shallow $\rightarrow$ deep	shallow $\rightarrow$ deep
Optimizer	SGD / Adam(W)	SGD / Adam(W)
Seeds	5–10 per configuration	
Compute	matched budgets	

The primary synthetic benchmark will be elementary cellular automata, following the controlled setting of Finzi et al. (Finzi et al., 2026). Their local transition rules provide tasks with different degrees of learnable structure while keeping the data-generating mechanism explicit.

For image data, we will additionally apply a fixed random permutation of pixels. Labels are unchanged, but the spatial arrangement exploited by a CNN is destroyed. The comparison therefore probes the interaction between data structure and architectural inductive bias.

The main control is matched compute rather than matched epochs.

Validation and potential application

For every task d , observer a , compute budget T , and seed s , we record the learning trajectory and derive

$$S_{\text{preq}}, \quad S_{\text{req}}, \quad \text{EDL}.$$

Different estimators will be compared by the ordering they assign to the same tasks, using Spearman correlation and Kendall's τ . Observer dependence will be quantified by

$$\Delta_{\text{arch}}(d, T) = S(d, \text{CNN}, T) - S(d, \text{MLP}, T).$$

Seed-to-seed variation will be reported explicitly; final test loss and sample efficiency will serve as external baselines.

Potential application: DiT with a sampler.

A natural extension is to study Diffusion Transformers together with their sampling procedures. The relevant computational object becomes

$$\text{noise} \longrightarrow \text{DiT} \longrightarrow \text{sampler} \longrightarrow \text{generated distribution}.$$

Different samplers use different numbers of model evaluations and follow different trajectories through the learned score field. After validating the methodology on lightweight observers, we can ask whether this compute-bounded perspective distinguishes sampling strategies beyond final sample quality alone.

Literature

- M. Finzi, S. Qiu, Y. Jiang, P. Izmailov, J. Zico Kolter, A. G. Wilson. *From Entropy to Epiplexity: Rethinking Information for Computationally Bounded Intelligence*. arXiv:2601.03220, 2026.
- S. Qiu, M. Finzi, Y. Zheng, K. Zhang, A. G. Wilson. *Requential Coding: Pushing the Limits of Model Compression with Self-Generated Training Data*. arXiv:2607.11883, 2026.
- E. Donoway, H. Joren, F. Roger, J. Leike. *Excess Description Length of Learning Generalizable Predictors*. arXiv:2601.04728, 2026.
- The project studies computationally bounded structural information through a controlled change of observer architecture, estimator, and compute budget.